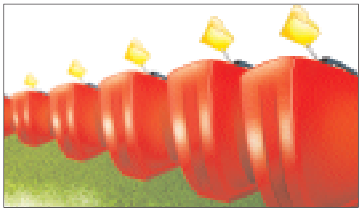

102 Multiple Redundancies

106 Online Services Vs. Software Suites

Tech Careers

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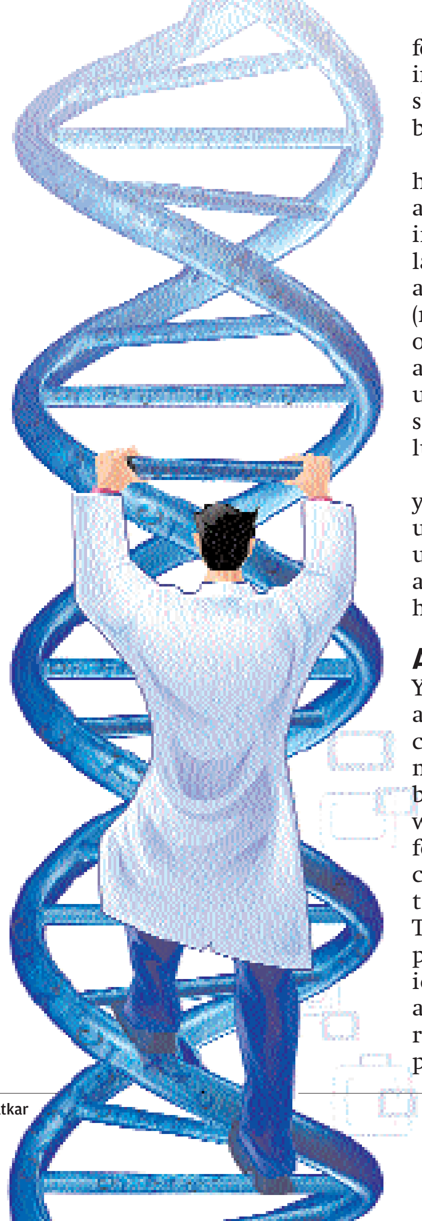
An offshoot of the Biotechnology industry, Bioinformatics is gathering pace like never before. It is still a niche industry, though

Asfaq Tapia

As the field of Biotechnology began to grow after 2001, it became increasingly clear that computers would play a big role in the future of commercial biology. Over the years, computers have played their part in decoding the human genome, drug and vaccine design, determining the bio-molecular structure of cells, and studying cell metabolism, among other things. This gradually evolved into a specialised industry called Bioinformatics.

Bioinformatics can, in simple terms, be defined as an interdisciplinary area involving biological, computer, and information sciences to manage, process, and understand large amounts of data, for instance from the sequencing of the human genome, or from large databases containing information gathered from plants and animals for use in discovering and developing new drugs. Mutation simulations and modelling of bio-molecules and biological systems is also done using the data. (Unfamiliar terms are explained in the box *Indicative Terms*, which might give you a flavour for what one deals with in this discipline.)

According to the Confederation of Indian Industries (CII), the Bioinformatics industry is still in its nascent stages, and is set to grow to $25 billion (Rs 1,00,000 crore) by 2011. Among other significant developments in the sector, the government of Karnataka along with ICICI invested, in 2001, Rs 5 crore to develop the Institute of Bioinformatics and Applied Biotechnology (IBAB) in Bangalore, to offer diploma courses and short-term training in bioinformatics and biotechnology.



Shrikrishna Patkar

The Bioinformatics industry today is heavily dependant on researchers who are able to work with custom software that involves data mining (sifting through large volumes of data to look for patterns and trends), downstream processing (referring to the separation or purification of biological products), and model analysis. Researchers comfortable with using Virtual Reality tools are also much sought-after in drug design and unicellular organism design.

This could be the industry for you if you are excited by the prospect of coming up with new cures for diseases, unlocking the secrets of life, and so on... and as you can expect, work here is highly research-oriented.

A Preview Of The Industry

You'll generally find bioinformatics applied in three broad cases: pharmaceutical companies that have a bioinformatics division; IT companies that have a bioinformatics wing—a small team working on specialised in-house projects for better data retrieval techniques; and companies that offer bioinformatics solutions to pharmaceutical companies. These range from research institutions, pharmaceutical companies, and chemical industries to the agriculture and allied industries. These companies recruit the bulk of the bioinformatics professionals today.